Test workflow

1. Upload test data
2. Explore measurements
3. Review engineering metrics
4. Investigate events
5. Export reports
6. Compare two tests

Reset analysis settings

Thermal Test Intelligence Center

Upload → Document → Visualize → Analyze → Review → Export | TTIC 1.0 · Stage 2D.3

Upload a test file

Sample Data.xlsx
1.8MB

Worksheet

Sheet1

Timestamp column

Timestamp

Observations	Numeric channels	Duration	Median sample interval
6,899	59	11.0 min	0.094 s

① Test Setup ② Visualization ③ Engineering Analytics ④ Event Intelligence ⑤ Reports & Export ⑥ Test Comparison

Event intelligence

Exploratory screening only. Detected events are observations, not pass/fail decisions unless an engineer supplies valid limits. All calculations use unsmoothed measurements.

Channels for event screening

TH1 × TH2 × TH3 × TH4 ×

Configure stability, sudden changes and limits

Stability window (seconds)
30.00

Sudden-change threshold (absolute units)
2.0000

Minimum readings per stability window
5

Maximum range for stable window (recorded units)
0.50

Maximum events shown
200

Use compatible units when applying the same tolerance across channels. Step transitions can legitimately trigger sudden-change events.

Maximum gap for grouping sudden changes (seconds)

1.00 − +

Optional acceptance limits

Configure limits separately for each channel. Blank limits mean no threshold check. Limits are inclusive; only readings outside them are flagged.

TH1	TH1 lower: No lower limit	TH1 upper: No upper limit
TH2	TH2 lower: No lower limit	TH2 upper: No upper limit
TH3	TH3 lower: No lower limit	TH3 upper: No upper limit
TH4	TH4 lower: No lower limit	TH4 upper: No upper limit

Sudden changes
250

Out-of-limit runs
0

Stable windows
12 / 84

Stable windows are non-overlapping, complete-duration bins with enough samples; the classification uses max − min, not standard deviation. An out-of-limit run is one or more consecutive readings outside the supplied bounds.

Stability windows

Channel	Windows	Stable	Largest_window_range	Stable %
TH1	21	3	101.9021	14.3
TH2	21	3	101.9021	14.3
TH3	21	2	101.7312	9.5
TH4	21	4	101.4577	19

Download stability windows (CSV)

Grouped sudden-change incidents

Grouped candidate incidents
15

Individual sudden changes
250

Grouping uses the gap between consecutive detected sudden changes, across selected channels. Start and end are detection times (not proven physical onset/recovery); duration is their difference. Single-detection incidents have a reported duration of 0 seconds. Out-of-limit runs remain separate in the detailed events table. Grouping does not imply shared cause or test failure.

Incident ID	Start time (s)	End time (s)	Duration (s)	Affected channels	Channel count	Sudden changes	Maximum absolute change	Time of maximum change (s)
1	7.249	7.719	0.47	TH1, TH2, TH3, TH4	4	7	3.897	7.719
2	11.188	13.454	2.266	TH1, TH2, TH3, TH4	4	18	12.9215	11.561
3	15.516	15.516	0	TH1, TH3, TH4	3	3	2.3929	15.516
4	19.469	19.469	0	TH3	1	1	2.3245	19.469
5	21.813	21.813	0	TH1, TH2, TH3	3	3	2.598	21.813
6	23.313	23.313	0	TH1, TH2	2	2	2.0852	23.313
7	25.281	25.281	0	TH1, TH2	2	2	2.1536	25.281
8	131.828	136.726	4.898	TH1, TH2, TH3, TH4	4	49	24.0313	132.511
9	251.563	254.531	2.968	TH3, TH4	2	14	10.5287	251.563
10	371.927	376.277	4.35	TH1, TH2, TH3, TH4	4	62	16.5792	372.694

Download grouped incidents (CSV)

Detected events

Channel	Event	Time (s)	Value	Previous value	Change	Delta time (s)	Configured limit
TH1	Sudden change	7.249	24.3975	26.5853	-2.1878	0.093	2
TH2	Sudden change	7.249	24.3291	26.7562	-2.427	0.093	2
TH1	Sudden change	7.439	21.8337	24.3975	-2.5638	0.095	2
TH2	Sudden change	7.439	21.9363	24.3291	-2.3929	0.095	2
TH4	Sudden change	7.439	25.3546	27.5424	-2.1878	0.095	2
TH3	Sudden change	7.719	21.15	24.7735	-3.6235	0.093	2
TH4	Sudden change	7.719	21.4577	25.3546	-3.897	0.093	2
TH1	Sudden change	11.188	16.6036	19.475	-2.8714	0.095	2
TH2	Sudden change	11.188	16.6378	19.4408	-2.8031	0.095	2
TH3	Sudden change	11.561	33.422	20.5005	12.9215	0.093	2

Showing the first 200 of 250 events. Download CSV for the complete list.

Download all detected events (CSV)

> View detected events on measurement traces

Automated observations (rule-based)

In the selected interval, TTIC screened 4 channel(s) and identified 250 adjacent-reading changes at or above the configured threshold and 0 runs outside supplied acceptance limits. 12 of 84 evaluable stability windows met the configured range tolerance. The 250 sudden changes form 15 time-grouped candidate incidents under the chosen grouping gap.

No acceptance limits were provided; zero out-of-limit events does not mean the test passed.